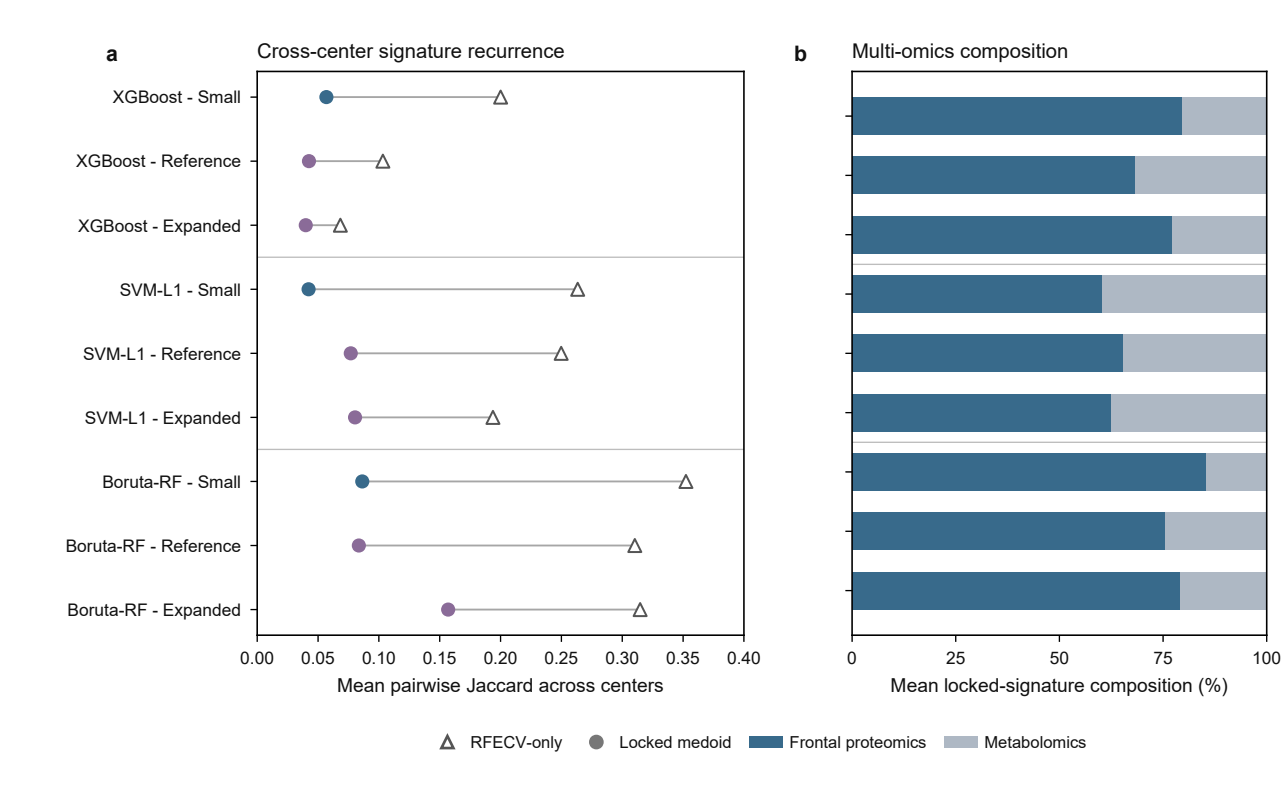

a
Cross-center signature recurrence
XGBoost - Small
XGBoost - Reference
XGBoost - Expanded
SVM-L1 - Small
SVM-L1 - Reference
SVM-L1 - Expanded
Boruta-RF - Small
Boruta-RF - Reference
Boruta-RF - Expanded
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Mean pairwise Jaccard across centers
b
Multi-omics composition
0
25
50
75
100
Mean locked-signature composition (%)
RFECV-only
Locked medoid
Frontal proteomics
Metabolomics